Reliable Annotation and Evaluation of Malaysian Malay-English Code-Mixed Sentiment Data: A Critical Review of an Unresolved Problem

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Abstract

Malaysian social media users mix English and Malay inside single messages, producing the contact variety known locally as Manglish. The studies reviewed here usually treat sentiment analysis of Malay-English mixed writing as a modelling problem. This review indicates that the more limiting problem originates earlier, in the sentiment labels themselves and in the evidence offered that those labels are sound. The review covers annotation practice and reliability evaluation for Malaysian Malay-English code-mixed sentiment data. Comparative benchmarking of sentiment classifiers is outside its scope. Three contributions follow. First, the annotation strategies used for low-resource code-mixed text are organised into three tiers, and each tier is shown to improve either label fidelity or scalability at the expense of the other. Second, the two causes of neutral-label skew are separated, one produced by the labelling tools and the other by genuine uncertainty among annotators. The four regional corpora are then examined against what a reliability claim requires, and none supplies it. Third, the class imbalance in the label distribution is shown to be the same condition under which Cohen's kappa becomes difficult to interpret. The review closes with a research agenda covering minimum reporting standards, agreement measurement under class imbalance, and the untested assumption that annotators of different types make different errors.

Keywords: *sentiment analysis, code-mixed text, Manglish, annotation reliability, inter-annotator agreement*

1. Introduction

Sentiment analysis, also known as opinion mining, is the automated task of deciding whether a piece of text expresses a positive, negative or neutral opinion (Poria et al., 2023). Organisations use it to measure public opinion at scale, in domains that include consumer products and services, healthcare, political elections and financial services (Nazir et al., 2022). The tools used for this task carry an assumption about their input. Joshi et al. (2020) show that only a small number of the world's languages are represented in current language technologies. Lexicons, pre-trained models and benchmark corpora are concentrated in those few languages (Joshi et al., 2020). A pre-trained model is one that has already learned general patterns from a large body of text, and is then adapted to a particular task such as sentiment analysis. Each of these resources is built for one language, so text in which two languages alternate within a single message falls outside the conditions they were designed for. Doğruöz et al. (2021) report that the effect on performance

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varies by task. Current models perform well on code-switched language identification and part-of-speech tagging, but only slightly better than chance on natural language inference (Doğruöz et al., 2021). Performance on code-switched text is therefore often lower than performance on monolingual text for the same task, though not uniformly so (Doğruöz et al., 2021; Winata et al., 2023). Malaysian social media text is one variety in which two languages alternate within a single message, and the two most often involved are English and Malay (AlAfnan et al., 2025; Treffers-Daller et al., 2022).

Malaysia is a multilingual and multi-ethnic country in which Coluzzi (2017), citing Ethnologue, records around 140 languages in use. Coluzzi (2017) also reports that code-mixing between a speaker's first languages and English or Malay is widespread in everyday communication. On social media, Malaysian users alternate between languages within a single message rather than writing in one language at a time (Treffers-Daller et al., 2022; AlAfnan et al., 2025). English and Malay are among the languages most frequently combined in Malaysian written communication (AlAfnan et al., 2025). Two local terms are used for such mixed writing, and they do not cover the same set of languages (AlAfnan et al., 2025; Romadhona et al., 2022). AlAfnan et al. (2025) describe *Manglish* as a hybrid form that combines Malay and English. Romadhona et al. (2022) describe *Bahasa Rojak* as a variety that combines English, Malay and Chinese. This review is concerned with Malay-English mixing, which *Manglish* denotes directly and *Bahasa Rojak* only in part.

How this mixed writing is regarded affects how it is processed, so this review states its position before turning to the technical difficulty. *Manglish* is sometimes described as a variety that weakens standard English and Malay rather than as a legitimate language variety in its own right (AlAfnan et al., 2025). Its structure, however, is not random. Treffers-Daller et al. (2022) analyse the classroom speech of two Malaysian teachers of English and find that Malay-English switching follows describable syntactic patterns. AlAfnan et al. (2025) report that speakers use *Manglish* for identifiable purposes, which include communicating efficiently, marking identity and building rapport.

The same regularity appears at the level of spelling, with Maskat and Rahman (2020) documenting the Malay negator *tidak* being written as *x*, and *tidak ada* being written as *takde*, *xde* or *tade*. Maskat and Rahman (2020) sort this variation into five categories and map two of them back to their standard forms, which indicates that the variation is systematic rather than arbitrary. A dictionary of standard Malay, such as the one curated by Dewan Bahasa dan Pustaka, does not list the non-standard forms. Tools that rely on such a dictionary therefore treat them as out-of-vocabulary items (Maskat and Rahman, 2020). Malay social media text has accordingly been described in earlier work as misspelled, out-of-vocabulary, ill-formed and noisy (Maskat and Rahman, 2020; Abu Bakar et al., 2020). That description reflects the reference standard being applied rather than a lack of care by the writer. The conventions Malaysian writers follow are consistent, but they differ from the conventions that standard natural language processing (NLP) tools were built to handle.

When a sentiment model performs poorly on Malay-English mixed text, one common explanation is architectural (Doğruöz et al., 2021; Winata et al., 2023). On this explanation the model handles code-switched input less effectively than monolingual input, and a larger or more regionally adapted model may perform better. The evidence for the explanation is mixed. Khanuja et al. (2020) report that multilingual models fine-tuned on code-switched data achieve the strongest scores among the systems evaluated on their benchmark. Adilazuarda et al. (2022) find that robustness to code-mixing varies with the embedded language, and attribute the variation to bias in the pre-training corpus. In contrast, Winata et al. (2021) find that pre-trained multilingual models do not necessarily guarantee high-quality representations of code-switched text. Conneau et al. (2020) describe a trade-off between positive transfer and capacity dilution, in which a model of fixed capacity that must accommodate more languages may produce representations of lower quality for each one. Doğruöz et al. (2021) identify a further limitation that is not architectural, in that the multilingual models in common use are trained mainly on monolingual corpora and may receive little exposure to code-switched text during pre-training. Model choice therefore accounts for part of the difficulty but not all of it.

Another source of difficulty does not originate in the model itself. It originates one step earlier, at the point where the labels are created. Supervised sentiment analysis requires a sentiment label for every message before a model can be trained or evaluated on it. Poria et al. (2023) note that this reliance on annotated data is often limiting. That label is assigned either by a human annotator or by an automatic one, and this step is called annotation. The labels it produces form the supervision signal used to train a model, and the labels in a test set are the standard against which the model's predictions are compared. Incorrect labels can affect both model training and subsequent evaluation. During training, a model may learn the patterns associated with incorrect labels, and Frénay and Verleysen (2014) report that the accuracy of predictions may decrease as a result. During evaluation, incorrect labels in a test set can bias the measurement of model performance, and Northcutt et al. (2021) show that they can change which of two models a benchmark ranks higher. Bowman and Dahl (2021) name annotation reliability as one of four conditions a sound evaluation benchmark must meet. An incorrect label can therefore have an effect at two separate stages, first when the model is trained and again when its performance is measured.

This review indicates that for Malaysian Malay-English code-mixed sentiment data, the more limiting problem at present is the annotation label rather than the model. A second and related problem is that the evidence these corpora offer for the soundness of their labels is limited. Three findings support that reading, and each is developed in the sections that follow. First, four Malay-English sentiment resources currently exist (Kasmuri and Basiron, 2019; Romadhona et al., 2022; Kong et al., 2023; Mohd Shamsuddin et al., 2024). None of them is at the same time large, balanced across sentiment classes, drawn from casual social media and validated by human annotators. Each of them secures some of these properties at the expense of the others. Second, the class imbalance in the largest of them appears to originate in the labelling method rather than in Malaysian discourse (Mohd Shamsuddin et al., 2024; Kong et al., 2023). Third, annotation reliability is rarely reported across

these four resources, and where a coefficient is given it was obtained for a narrower task than the corpus is used to support (Kasmuri and Basiron, 2019; Romadhona et al., 2022; Kong et al., 2023; Mohd Shamsuddin et al., 2024).

The second and third findings describe one condition rather than two. Reliability is normally demonstrated by having independent annotators label the same text and computing a chance-corrected agreement coefficient, the most widely used of which is Cohen's kappa (κ) (Cohen, 1960; Artstein and Poesio, 2008). Cohen's κ becomes difficult to interpret when one category dominates the label distribution, because the agreement expected by chance rises with that skew (Feinstein and Cicchetti, 1990; Byrt et al., 1993). Di Eugenio and Glass (2004) call this the prevalence problem. The skew that the labelling methods produce is therefore also the condition under which the statistic used to check the labels stops being informative, so correcting either weakness alone leaves the other in place. A 2024 review of the field still reports a lack of comprehensive datasets and annotated resources for code-mixed Malay text (Selvaraju et al., 2024).

This review examines one question: why reliable annotation and evaluation of Malaysian Malay-English code-mixed sentiment data remain unresolved. Linguistic description, model architecture and application domain are discussed only where they are relevant to that question. Two related topics fall outside the scope. The first is comparative benchmarking of sentiment classifiers, and the second is sentiment analysis as a general field. Both topics have already been surveyed in detail (Poria et al., 2023; Nazir et al., 2022; Selvaraju et al., 2024). This review does not propose, implement or evaluate a system, and it does not report the author's own experimental work. Its contribution is a diagnosis of why the problem remains unresolved, together with a set of directions for future research.

This review makes three contributions:

1. It organises the annotation strategies used for low-resource code-mixed text into three tiers, which are manual, semi-automated and fully automated labelling, and shows that each tier gains either label fidelity or scalability at the expense of the other.
2. It distinguishes the two causes of neutral-label skew, one produced by the labelling tools themselves and the other by genuine uncertainty among annotators, and it examines what each of the four regional corpora reports about its own reliability.
3. It shows how the linguistic, annotation and measurement challenges compound one another rather than merely co-occurring, and sets out a research agenda that follows from that interaction.

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